ASTANA IT UNIVERSITY

«Approved»
by the Director of SAI and DS
Sergaziev M.Zh.
«29» 08 2025

Syllabus
Academic Year 2025 – 2026

<table>
<tr><th colspan="2">1. General information</th></tr>
<tr><td>Course Code</td><td>Math 1201</td></tr>
<tr><td>Course Title</td><td>Calculus I</td></tr>
<tr><td>Degree Cycle (Level)/ Major / Relation to curriculum</td><td>Bachelor's Degree
SE, IOT, MCS, CS, ST
Computer Science</td></tr>
<tr><td>Year, trimester</td><td>1st year, 1</td></tr>
<tr><td>Language of Instruction:</td><td>English</td></tr>
<tr><td>Lecturer(s)/ Instructors/ Instructor Contact Information</td><td>Mira Rakhimzhanova, PhD, Assistant-Professor
m.rahimzhanova@astanait.edu.kz
Office: C1.1.337
Serik Kudaibergenov, PhD Candidate of Mathematics , Senior-Lecturer
Serik.Kudaibergenov@astanait.edu.kz
Office: C1.1.337</td></tr>
<tr><td>Number of Credits</td><td>5</td></tr>
<tr><td>Workload of course components and credits per trimester</td><td>
<table>
<tr><th rowspan="2">ECTS credits</th><th colspan="3">Contact hours</th><th rowspan="2">IASS</th><th rowspan="2">SIS</th><th rowspan="2">Total hours</th></tr>
<tr><th>Lectures</th><th>Practice sessions</th><th>Laboratory work</th></tr>
<tr><td>5</td><td>20</td><td>20</td><td></td><td>10</td><td>90</td><td>150</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>
</td></tr>
<tr><td>Prerequisites</td><td>School math courses</td></tr>
<tr><td>Post requisites</td><td>Calculus II, Theory of Probability and Statistics</td></tr>
<tr><th colspan="2">2. Goals, objectives and learning outcomes of the course</th></tr>
<tr><td>1. Course Description</td><td>Calculus is a foundational course at Astana IT University; it plays an important role in understanding science, engineering, economics, and computer science, among other disciplines. This Calculus-1 course covers the differentiation and integration of functions of one variable with applications.</td></tr>
<tr><td>2. Course Learning Goals</td><td>• After completing this course, students should have developed a clear understanding of the fundamental concepts of single-variable calculus and a range of skills that allow them to work effectively with them. The main goal is to be able to apply the tools mentioned above to problems in post-requisite courses.</td></tr>
<tr><td>3. Course Learning Outcomes</td><td>Students should study the following concepts:
• Sets and functions of one variable
• Limits of sequences and their properties
• Continuity and properties of continuous functions
• Derivatives and their applications
• Antiderivatives and methods to evaluate them
• Definite integrals in 1D and 2D cases
• Improper integrals</td></tr>
</table>

1. Methods/forms of teaching	4. Methods/forms of teaching to apply the tools mentioned above to problems in post-requisite courses. Lectures - cover theoretical foundations and their practical applications. Practical classes - focus on applying theories through problem- solving and implementation. Assignments - individual work for each topic, including coding and manual solution. Discussion - collaborative analysis of concepts, ideas, and open questions
2. Coursework and Grading Scheme	5. Coursework and Grading Scheme Grading schemes involve exams (midterms, endterm, final exam), quizzes, and homework assignments.
3. Academic Integrity	6. Academic Integrity Follow the Rules of Academic Integrity approved by the Scientific Council of LLP "Astana IT University"
4. Learning resources:	Basic Literature: 1. Lecture presentations; 2. Thomas' Calculus. By George b. Thomas, revised by J.Hass, C.Heil, M.D.Weir, Pearson Publishing Company. 14n edition 3. George b. Thomas, Jr., Ross L. Finney, Calculus and Analytic Geometry. Part II. Addison-Wesley Publishing Company. 9 th edition. Supplementary literature: 4. G. N. Berman, A collection of problems on a course of Mathematical Analysis 5. Г.М.Фихтенгольц. Основы математического анализа, Т.1, Изд-е 9-ое, Изд. Лань – 2008. – 448 с. 6. Ибрашев Х.И., Еркеғұлов Ш.Т. Математикалық анализ курсы. - Алматы, 1970. 7. Темірғалиев Н.Т. Математикалық анализ. 1 бөлім. - Алматы: Мектеп, 1987.
5. Technology employed:	Assigned reading materials and presentations should be read before the class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and/or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu .kz).
6. Course Policies	**Course and University policies include:** Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally, students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse, the instructor has the right to mark him as "not graded". In such a case, a student is not admitted to the exam and automatically fails the course. In cases when a student misses a class session due to valid reasons (is excused by the instructor or the dean's office) he/she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he/she has to study material provided under the course on their own. The course instructor might **Preparation for Class**: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments.

	Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes. **Being late on class:** When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor. **Attestation I and II:** Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course. **Home work / Assignments**: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures. **Late submissions:** Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale. *In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.* **Final exam:** Project (report) and presentation (slides). At the completion of this course, each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit to the LMS Moodle the Written Report of Final project and the Slide-Presentation 2-3 days before the Final exam day. **How will you get feedback after every Exam?** After posting your grades for Exams (it should happen within 3-5 working days after the exam), I will briefly discuss the most difficult questions of the test in the class and will give you general recommendations. After that, if you want to take a look at your answer sheet, you can come to office hours (only in-person, I do not provide online consultation for exams). I will not allow you to take a photo of it or to take it with you. If you want to not only take a look but also receive clarification and ask some questions, I will give you a consultation, but only if you follow my general recommendations first. If you did not follow my general recommendations, I may suggest postponing the discussion of your exam results until you implement the general recommendations I gave in class. Please be respectful when discussing results and also refrain from the emotional blackmail (requests to raise the grade, to give additional assignments, etc, see this course policy), otherwise I will stop the consultation immediately. Also note, that you can look at your answer sheet and ask for a discussion of your results only during the period of Exam weeks. **Laptops and mobile devices** can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. **Online lessons** can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu .kz) as a Learning Management System. **Behaviour during classes:** Please do not talk and distract lecturer and other students during classes unless the instructor explicitly gives permission to talk. Please do not disturb the lecturer or students in any way during the classes. Violating these rules will lead to a warning and in case of a second

	violation will lead to complaint to Ethics Committee, which may decide to give the student an official warning or exclude the student from the course. **Emotional blackmail:** Please refrain from the emotional blackmail towards the lecturer. It is against the University's Code of Conduct and potentially a subject of a complaint to the Ethics Committee. Examples of emotional blackmail include: * begging lecturer to increase your grade due to the possibility of losing a scholarship, stipend, going to academic mobility etc.; •asking lecturer to "understand you, your position", "financial difficulties" etc.; * asking to add additional points because "I only need 3 points to pass"; * asking to give you an additional assignment to increase your grade. Please remember instructor do not "assign" grades. Students earn grades. A lecturer cannot give a student a grade she/he did not earn. **Cheating and plagiarism** are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. **Academic Conduct Policies of the university:** The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu .kz). **Academic Conduct Policies of the university:** The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). **Contacting the Course instructor**: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.
7. Course Schedule	See Section 3.2-3.4
8. Syllabus Inclusion Statements	• Listening actively - respecting others when they are talking • Speaking from one's own experiences instead of generalizing ("I" instead of "they," "we," and "you") • Participating to the fullest of one's ability - community growth depends on the inclusion of every individual voice • • Helping self and peers to become more self-reflective • Confidentiality - respecting the confidentiality of personal disclosure Supporting one another if someone needs clarification, a chance to speak, a chance to quietly gather thoughts, or time to think/reflect before responding to a question

3.1 Abbreviations

#	Abbreviation	Meaning
1	IASS	Instructor assisted student study - Instructors assist students in achieving success by delivering support, feedback, and resources. This support might

		take the form of clarifying difficult topics, recommending additional materials, or providing individualized guidance.
2	SIS	Student independent study
3	IW	Individual work
4	GW	Group work
5	PA	Practical assignment
6	LW	Laboratory work
7	MCQ	Multiple choice quiz
8	PBT	Paper-based test

NB: You may introduce abbreviations that are used in the syllabus

3.2 Course Plan

Week No	Course Topic	Lectures (h/w)	Practice sessions (h/w)	Lab. Sessions (h/w)	IASS (h/w)	SIS (h/w)
1	Sets and Functions.	3	2		1	9
2	Sequences and Limits.	3	2		1	9
3	Limits and Continuity.	3	2		1	9
4	Derivatives.	3	2		1	9
5	Application of derivatives.	3	1		1	9
6	Indefinite integrals.		1		1	
7	Methods of Integration	3	2		1	9
8	Definite integrals.	3	2		1	9
9	Application of Integrals.	3	2		1	9
10	Complex numbers. Polar coordinates.	3	2		1	9
		3	1		1	9
	Total:	**30**	**20**		10	**90**

3.3 Detailed course plan

Week No	Lecture Topics with Learning Resources	Tasks and Activities for Practice Sessions with Recommended Reading Resources
1	**Sets and Functions.** Sets and Operations on Sets. Functions and their graphs. Shifts, and Scaling graphs. Elementary functions, the transcendental functions. Inverse Functions. The composition of functions.	Books provided in the Learning resources
2	**Sequences and Limits.** Sequences converging to a limit. Properties of convergent sequences. Monotone sequences. Limits of function values. Definition of a Limit. The Sandwich theorem.	Assignment 1 Books provided in the Learning resources

3	**Limits and Continuity.** Continuous functions. One-Sided limits. Limits involving infinity. Relative Rates of growth. (Table of equivalent functions).	Books provided in the Learning resources
4	**Derivatives**. The definition of derivative. Rules. Derivatives of trigonometric and transcendental functions. The Chain rule. Second and higher order derivatives. Implicit differentiation. Linearization and differentials. Derivative of the parametric functions, Tangents and normal of lines.	Assignment 2 Books provided in the Learning resources
5	**Application of derivatives**. Maxima and Minima. The mean value theorem. Graphing by the first and the second derivatives. Asymptotes and Dominant terms. Graphing Rational Functions. Related rates of change. Optimization. Indeterminate forms and L'Hopital's Rule theorem, Midterm Exam.	Midterm Exam Books provided in the Learning resources
6	**Integrals.** Antiderivatives. Indefinite integrals. Basic Integration Formulas. Substitutions in indefinite integrals.	Books provided in the Learning resources
7	**Integrals.** Integration by parts. Special cases of integration. Integrals for Trigonometric and Rational Functions, Partial Fractions.	Assignment 3 Books provided in the Learning resources
8	**Definite integrals.** Area and estimating with finite sums. Sigma notation and limits of finite sums. The fundamental theorem of Calculus. Substitution in definite integrals. Improper integral.	Books provided in the Learning resources
9	**Application of Integrals.** The area between curves. Volumes. Arc Length. Areas of surfaces of revolution.	Assignment 4 Books provided in the Learning resources
10	**Complex numbers.** Complex numbers and operations with them. Polar coordinates. Graphing polar coordinate equations. Integration in polar coordinates. Area and Lengths in polar coordinates.	Endterm Exam Books provided in the Learning resources

3.4 List of assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	**2**	**3**	**4**	**5**
1	1.1 Functions, their domain, range, and graphs. Shifts, and Scaling graphs. Inverse Functions. The composition of functions.	9	[2]: Ch. 1.1 – 1.3 and Ch. 7.1-7.3, Appendix A.1	Electronic version or hard cory of the assignment
2	1.2 Sequences converging to a limit. Limits of convergent sequences and functions by definition.	9	[2]: Ch. 8, Appendix 10.1, [3], Ch1	Electronic version or hard cory of the assignment
3	2.1 Computing Limits of Function values and applying Theorems. Limits involving infinity. Estimate relative Rates of growth.	9	[2]: Ch. 2 and 7	Electronic version or hard cory of the assignment
4	2.2 Compute derivatives by definition and	9	[2]: Ch. 3	Electronic version

	applying rules. Derivatives of trigonometric and transcendental functions. The Chain rule. Second and higher order derivatives. Implicit differentiation. Linearization and differentials. Derivative of the parametric functions, find tangents and normal of lines.		and 7	or hard cory of the assignment
5	3.1 Find Maxima and Minima. Graphing by the first and the second derivatives. Find Asymptotes and Dominant terms. Apply L'Hopital's Rule and compute limits for Indeterminate forms	9	[2]: Ch. 4,7	Electronic version or hard cory of the assignment
6	3.2 Compute antiderivatives. Use Basic Integration Formulas. Make Substitutions in indefinite integrals	9	[2]: Ch. 5, 8	Electronic version or hard cory of the assignment
7	3.3 Apply integration by part. Study special cases of integration for Rational and trigonometric Functions	9	[2]: Ch. 5, 8	Electronic version or hard cory of the assignment
8	4.1 Compute the area between curves and Areas of surfaces of revolution.	9	[2]: Ch.5, 8	Electronic version or hard cory of the assignment
9	4.2 Compute volumes and Arc Length.	9	[2] Ch. 6, 8	Electronic version or hard cory of the assignment
10	4.3 Exercise with complex numbers and polar coordinates	9	[2]: Ch. 11.3 – 11.5, Appendix A.7	Electronic version or hard cory of the assignment

3.5 For instructor assisted student study (IASS) schedule and policy follow your course LMS Moodle.

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1st attestation	**Assignments**:** assignment 1, assignment 2 Quiz 1 **Mid Term**	**60** 20 20 20 **40**	100

2nd attestation	**Assignments**:** assignment 3, assignment 4, Quiz 2 **End Term**	**60** 20 20 20 **40**	100
Final exam*	Written Exam		100
Total	**0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final**		**100**

** The complete title of the assignment must be clearly stated.*
*** The number of assignments may vary depending on the course content and is determined by the course syllabus.*
**** Description of iROS or Social GPA elements, if applicable*

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	GPA	Percentage	**Description**
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	30-49	Fail
F	0	0-29	

Course Assessment Rubric for Student Performance Evaluation

Learning outcomes	Indicators	Excellent (90-100)	Good (70-89)	Satisfactory (50-69)	Fail 0-49
Understand concepts related to sets, limits, continuity, derivatives, and basic integrals	Defines sets, functions, domain/range Explains limits and continuity with examples Understands derivative as the rate of change Explains the meaning of definite/indefinite integrals	Provides clear, precise definitions and explanations with examples; no conceptual errors	Provides mostly correct definitions, with minor inaccuracies in examples	Gives partial Or vague definitions, with gaps in understanding	Unable to define key concepts or demonstrate serious misconceptions
Apply and	Evaluates limits	Solves all	Solves most	Solves only	Unable to

work euth this concepts numerically, graphically and analitically	and continuity Computes derivatives using standard rules Sketches curves using derivatives (extrema, concavity, asymptotes) Computes definite and indefinite integrals	problems accurately across algebraic, numerical, and graphical forms	problems correctly, with occasional minor errors	basic problems, with frequent mistakes in reasoning	solve problems; incorrect or missing solutions
3. Apply the tools mentioned above to problems in post- requisite courses.	Solves optimization and related rates problems Applies integrals to compute areas and volumes Transfers knowledge to physics/economics/engineering problems	Correctly sets up and solves all application problems, including non-routine tasks	Correctly solves standard problems; struggles with complex/unfamiliar ones	Solves only simple application problems, weak in interpretation	Cannot apply concepts to applications; fails to set up problems

5. Course assessment methodology

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge.
- **Final assessment** is a Written Exam.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Ramazanova Zamart	Senior-Lecturer	27.08.2025	
Aizada Kadyr	Teacher	27.08.2025	